

TUSCALOOSA, AL, USA

WMO: 722286

Lat: 33.212N

Lon: 87.616W

Elev: 150

StdP: 14.62

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	26.5	7.7	8.2	29.7	12.0	10.2	34.7	19.6	47.4	18.1	48.8	4.8	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	97.3	75.4	94.7	75.7	92.6	75.4	79.7	89.5	78.7	88.6	77.9	87.7	7.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.8	83.0	76.2	137.4	82.5	75.2	133.0	81.9	43.2	89.4	42.2	88.8	41.3	87.9	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.1	15.1	12.8		DB	16.0	100.1	4.3	3.9	12.9	102.9	10.4	105.2	8.0	107.3	4.8	110.2
				WB	14.5	81.6	4.4	1.6	11.3	82.7	8.8	83.7	6.3	84.6	3.1	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	45.1	49.7	56.8	64.5	72.4	79.3	81.7	81.5	76.1	65.3	54.5	48.3
	DBStd	14.94	10.27	9.81	8.94	7.00	5.99	3.83	3.04	3.45	5.69	8.07	8.47	9.36
	HDD50	580	222	117	39	2	0	0	0	0	0	2	49	149
	HDD65	2396	619	438	277	94	16	0	0	0	4	98	328	522
	CDD50	5941	71	108	251	436	695	879	983	976	784	476	185	97
	CDD65	2279	3	8	24	77	245	429	518	511	338	107	14	5
	CDH74	22395	5	36	228	769	2310	4305	5303	5196	3165	980	83	15
	CDH80	9946	0	2	31	162	868	2052	2580	2573	1377	289	10	2
Wind	WSAvg	5.0	6.0	6.2	6.3	5.9	5.0	4.3	4.1	3.8	4.3	4.3	4.9	5.5
Precipitation	PrecAvg	52.6	5.3	5.1	5.7	4.9	4.0	4.3	4.8	3.6	3.2	3.2	4.4	4.7
	PrecMax	75.4	9.6	14.5	17.7	15.0	11.1	12.0	12.3	7.4	8.7	9.1	11.4	10.4
	PrecMin	35.4	1.1	1.3	1.2	0.4	0.4	0.4	0.9	0.7	0.8	0.1	1.0	1.0
	PrecStd	9.6	2.0	2.8	3.2	2.9	2.6	2.5	2.6	1.8	2.2	2.3	2.2	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.1	78.8	83.6	87.4	93.7	98.9	99.5	100.4	96.8	90.3	81.4	75.5
		MCWB	63.2	66.5	64.7	70.4	73.4	74.0	75.3	75.4	73.5	72.2	67.4	66.4
	2%	DB	70.3	73.6	79.7	84.1	90.2	95.3	96.6	97.4	93.2	86.3	76.1	71.7
		MCWB	62.5	63.4	63.8	68.0	72.4	74.8	76.7	75.8	73.9	70.0	64.4	64.7
	5%	DB	66.1	70.3	76.2	81.7	87.9	92.7	93.4	94.1	90.7	82.9	72.9	67.8
		MCWB	60.5	61.1	62.3	66.6	71.4	74.8	76.6	75.8	73.5	68.3	63.3	62.3
	10%	DB	62.8	66.2	72.6	79.1	85.0	90.3	91.3	91.3	88.1	80.1	70.0	64.0
		MCWB	56.9	58.4	60.1	64.9	70.5	74.5	76.5	76.0	72.9	67.4	62.0	58.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	69.0	69.4	73.2	77.6	79.8	81.1	80.9	78.3	76.6	72.1	69.6
		MCDB	71.7	74.5	76.3	82.2	86.9	90.5	90.0	90.4	88.1	83.4	75.4	72.9
	2%	WB	64.3	66.3	66.9	71.0	75.3	78.3	79.9	79.5	77.1	74.4	69.1	66.2
		MCDB	67.6	72.0	74.3	79.9	85.0	88.8	89.6	89.5	86.9	82.1	72.8	69.7
	5%	WB	61.6	63.8	65.0	69.2	73.8	77.4	78.9	78.5	76.0	72.4	66.3	63.4
		MCDB	66.1	68.2	72.7	77.4	83.1	87.8	88.8	88.2	85.3	78.8	70.9	66.7
	10%	WB	58.0	60.0	62.9	67.5	72.5	76.4	78.0	77.7	74.9	70.2	63.4	59.9
		MCDB	62.3	64.7	70.6	75.2	81.7	85.9	87.8	87.2	83.5	76.1	67.4	63.6
Mean Daily Temperature Range	5% DB	MDBR	20.7	21.6	23.0	24.0	22.0	20.6	19.6	20.0	21.5	23.4	23.3	20.4
		MCDBR	22.7	24.4	26.5	25.9	24.5	24.6	22.8	24.1	23.8	26.1	25.5	22.2
		MCWBR	16.0	15.1	13.5	12.0	9.2	7.9	7.2	7.1	8.0	11.4	14.7	15.5
	5% WB	MCDBR	18.9	19.7	21.2	20.9	19.6	19.7	19.4	19.2	18.9	18.4	18.4	18.4
		MCWBR	15.1	14.6	12.4	11.6	8.5	7.7	7.4	7.2	7.3	9.7	13.3	15.2
Clear-Sky Solar Irradiance	taub	0.324	0.335	0.365	0.401	0.450	0.482	0.533	0.532	0.443	0.373	0.347	0.324	
	taud	2.519	2.482	2.418	2.304	2.214	2.156	2.035	2.031	2.270	2.452	2.465	2.534	
	Ebn at Noon	283	291	289	282	268	258	244	242	261	274	272	277	
	Edh at Noon	27	31	35	41	46	48	54	54	41	31	28	25	
All-Sky Solar Radiation	RadAvg	842	1009	1376	1745	1921	1994	1916	1762	1558	1291	954	719	
	RadStd	68	139	128	136	142	155	120	110	134	159	100	72	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+257	+185

Nomenclature: See separate page